

Automated Network Request Management

A ServiceNow Service Catalog solution

Project	Automated Network Request Management
Platform	ServiceNow - Service Catalog
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Date	[Month Year]

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1. INTRODUCTION

1.1 Project Overview

Automated Network Request Management is a ServiceNow solution that lets employees raise network-related requests through one structured Service Catalog item called **Network Request**. The item sits in the Service Catalog under the category *Network Standard Changes* and is owned by the System Administrator.

The form is built from reusable parts: seven catalog variables, a **Requester Information** variable set (four variables, including 'opened on behalf of'), a Catalog UI Policy that shows or hides the ID field, and Catalog Client Scripts that populate and validate fields in the browser. Submitted requests are handled by the ServiceNow process engine. A custom table, `u_database_tables`, supports the project's data.

1.2 Purpose

- Replace email- and phone-based network requests with one consistent, trackable form.
- Collect complete and correct information up front through mandatory fields and conditional logic.
- Reduce manual effort and turnaround time for standard network changes.
- Give requesters and the network team a single record to follow from submission to closure.

2. IDEATION PHASE

2.1 Problem Statement

Network standard changes (access, connectivity and similar requests) are often raised through emails, chats or unstructured tickets. Important details are missing, requests are routed to the wrong people, and nobody can easily see their status. This causes delays, repeated follow-ups and inconsistent handling.

No.	I am...	I'm trying to...	But...	Because...	Which makes me feel...
PS-1	An employee needing a network change	Raise a request and know it will be handled	I don't know what details are needed or who to send it to	Requests are informal and spread across email and chat	Frustrated and uncertain
PS-2	A network team member	Process requests quickly and correctly	Requests arrive incomplete and in different formats	There is no standard intake form	Overloaded and slowed down

2.2 Empathy Map Canvas

Empathy map for the requester (employee raising a network change):

SAYS "Where do I send my network request?" "Has anyone looked at my request yet?"	THINKS I may forget a detail and get it bounced back. It should be as easy as ordering a laptop.
DOES Emails several people, follows up repeatedly, re-sends details.	FEELS Frustrated by delays, anxious about status, unsure the request was understood.
PAINS Missing information, no status visibility, repeated follow-ups, inconsistent responses.	GAINS One clear form, mandatory checks, instant confirmation, trackable request number, faster fulfilment.

2.3 Brainstorming

Ideas generated by the team were grouped and then prioritised. Ideas marked **Selected** were implemented.

Theme	Idea	Decision
Intake	Single Service Catalog item for all network requests	Selected
Intake	Reusable 'Requester Information' variable set shared across items	Selected
Validation	UI Policy to show/hide the ID field based on the request context	Selected
Validation	Client scripts to auto-populate fields and enforce mandatory attachments	Selected
Process	Process engine to route the request after submission	Selected
Data	Custom table to store supporting project data	Selected
Process	Multi-level approvals and automated fulfilment with Flow Designer	Future scope
Experience	Email / SMS notifications and status dashboards	Future scope
Integration	Direct integration with network devices / CMDB	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Awareness	Request	Validation	Fulfilment	Closure
Doing	Realises a network change is needed	Opens Service Catalog, selects Network Request	Fills the form; the UI policy and scripts guide input	Request moves through the process engine	Receives confirmation and closes the request
Thinking	Who handles this?	Is this the right item?	Did I give everything needed?	When will this be done?	Was it done correctly?
Feeling	Uncertain	Confident - one clear place	Reassured by instant checks	Informed via request number	Satisfied
Pain points	No known intake route	Too many options	Missing details found late	No status visibility	Reopening for mistakes
Opportunities	Promote the catalog item	Clear category: Network Standard Changes	Mandatory fields, conditional fields	Notifications and tracking	Feedback survey

3.2 Solution Requirement

Functional requirements

ID	Requirement	Priority
FR-1	Provide a 'Network Request' item in the Service Catalog under Network Standard Changes	High
FR-2	Capture requester details through the Requester Information variable set (incl. 'opened on behalf of')	High
FR-3	Collect request details through seven catalog variables	High
FR-4	Show or hide the ID field automatically using a Catalog UI Policy	Medium
FR-5	Populate fields and enforce mandatory attachments using Catalog Client Scripts	Medium
FR-6	Route submitted requests through the process engine	High
FR-7	Allow the item to be tested using 'Try It'	Low

Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	Form is simple, with clear labels and conditional fields.
NFR-2	Security	Access limited by roles and ACLs; item availability can be restricted.
NFR-3	Reliability	Runs on the managed ServiceNow platform with its availability.
NFR-4	Performance	Form loads and validates in the browser without noticeable delay.
NFR-5	Maintainability	Reusable variable set and low-code configuration make changes easy.

3.3 Data Flow Diagram

Level-1 data flow of the Network Request process. Rectangles with a coloured fill are external entities; D1 and D2 are data stores.

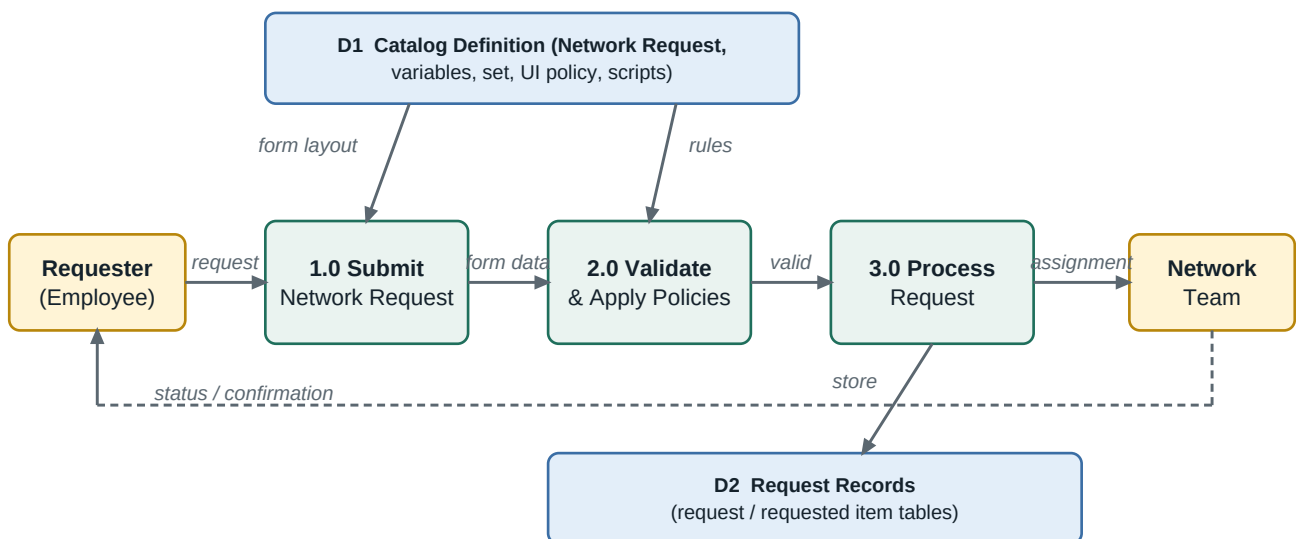


Figure 1 - Level-1 Data Flow Diagram

3.4 Technology Stack

Component	Technology / Tool
Platform	ServiceNow (developer instance)
Module	Service Catalog (Catalog Definitions, Catalog Items, Variable Sets)
Front-end logic	Catalog UI Policies, Catalog Client Scripts (JavaScript, onLoad / onChange / onSubmit)
Back-end logic	Process engine, platform business logic
Database	ServiceNow relational tables, including custom table u_database_tables
Access	Web browser; role-based security and ACLs
Testing	Catalog 'Try It', manual functional tests

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer segment	Employees who need network changes; network / IT team who fulfil them
Problem	Unstructured, incomplete and untracked network requests
Existing alternatives	Email, chat messages, generic IT tickets
Our solution	A single, validated Service Catalog item with automatic routing
Triggers to act	Repeated rework, delayed network changes, lack of visibility
Outcome	Complete requests on first submission, faster handling, a trackable request number

4.2 Proposed Solution

Aspect	Description
Idea	Provide a self-service Network Request item in ServiceNow that guides the requester and feeds the process engine.
Novelty	Reusable variable set + UI policy + client scripts make the form smart without custom application code.
Social impact	Saves staff time, reduces errors and delays in network services.
Business model	Internal IT service; value comes from reduced effort and faster turnaround.
Scalability	New request types can be added as catalog items reusing the same variable set and patterns.

4.3 Solution Architecture

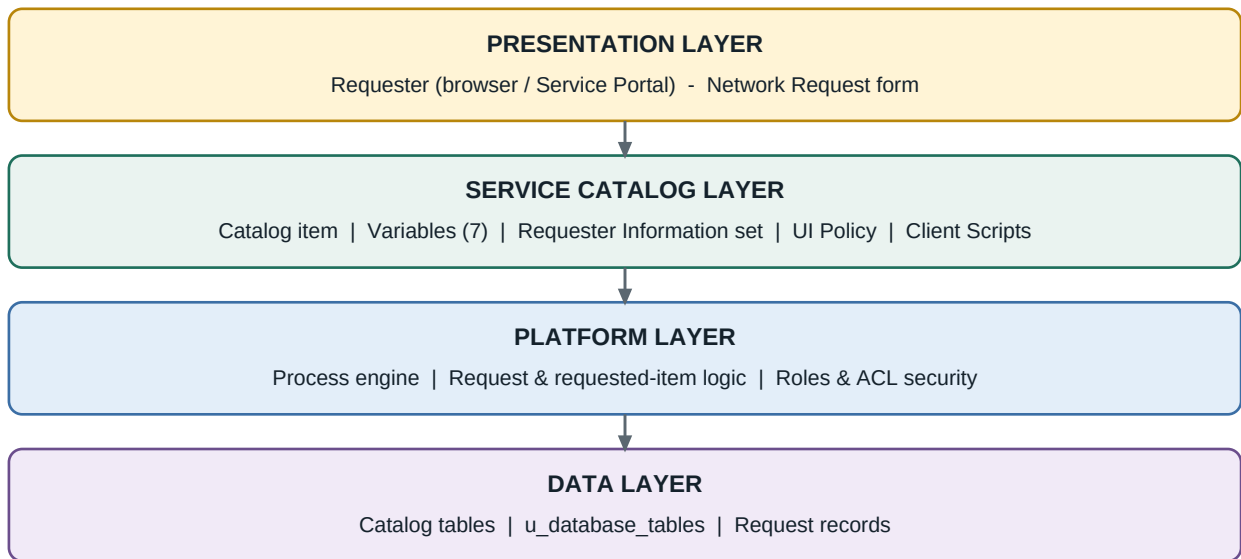


Figure 2 - Solution architecture

The requester interacts only with the form. Client-side policies and scripts validate input in the browser; on submission the platform's process engine creates and routes the request, and all records are stored in ServiceNow tables.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was organised into four sprints. Dates are placeholders - replace them with your actual schedule.

Sprint	Scope / User story	Points	Duration	Status
Sprint 1	Requirement gathering, problem definition, create custom table u_database_tables	5	[dd-mm] to [dd-mm]	Completed
Sprint 2	Create the Network Request catalog item, category and seven variables	8	[dd-mm] to [dd-mm]	Completed
Sprint 3	Build Requester Information variable set, UI Policy and Catalog Client Scripts	8	[dd-mm] to [dd-mm]	Completed
Sprint 4	Process engine setup, testing with Try It, documentation and demo	6	[dd-mm] to [dd-mm]	Completed

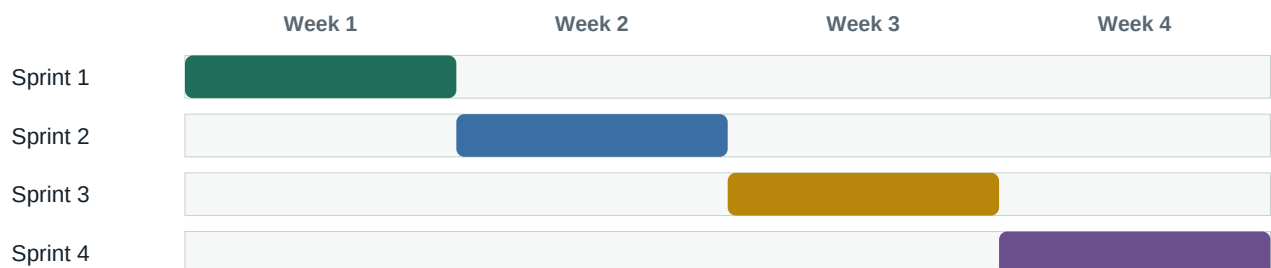


Figure 3 - Sprint timeline (illustrative, one sprint per week)

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing was carried out using the catalog item's **Try It** option and by opening the form as an end user. The test cases below describe what was verified. Record the actual outcome in the Result column.

ID	Test case	Expected result	Result
TC-1	Open Network Request from Maintain Items	Item opens with Service Catalog and Network Standard Changes category	[Pass]
TC-2	Open the form with 'Try It'	Form shows Requester Information set and variables	[Pass]
TC-3	Use 'opened on behalf of' reference	User can be selected and the value is kept	[Pass]
TC-4	Trigger the ID field UI policy	ID field shows or hides as defined (On load, Reverse if false)	[Pass]
TC-5	Leave a mandatory variable empty and submit	Submission is blocked with a message	[Pass]
TC-6	Change a field that has an onChange script	Dependent field is populated or hidden automatically	[Pass]
TC-7	Submit a valid request	Request record is created and a request number is shown	[Pass]

Performance aspect	Observation approach
Form load time	Open the item repeatedly; note time until the form is usable. [Record value]
Client script response	Check that onChange behaviour is instant. [Record value]
Submission time	Time from clicking Submit to confirmation. [Record value]

7. RESULTS

7.1 Output Screenshots

The screenshot shows the ServiceNow interface for editing a catalog item. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Admin', and a search bar. The main header shows 'Catalog Items' with a search icon. Below this, the breadcrumb 'Catalog Item > Network Request' is visible. The form itself is titled 'Build and modify items faster with the improved Catalog Builder.' and contains several fields: 'Name' (Network Request), 'Application' (Global), 'Active' (checked), 'Roles' (empty), 'Fulfillment automation level' (Unspecified), 'State' (None), 'Checked out' (None), 'Owner' (System Administrator), and 'Secondary owners' (empty). At the bottom, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Short description' field is filled with 'Network Services Request'. The URL at the bottom of the page is 'https://dev393606.service-now.com/sc_cat_item.do?sys_id=f782bcb79323c49...'.

Figure 4 - Network Request catalog item (Service Catalog, Network Standard Changes).

servicenow All Favorites History : Catalog Item - Network Request

Copy Try It Update Edit in Catalog Builder Delete

Related Links
[Item Diagnostic](#)
[Show VA render type](#)
[Run Point Scan](#)

Variables (7) Variable Sets (1) **Catalog UI Policies (1)** Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions

Related Articles Related Catalog Items **Assigned Topics**

Order Search Actions on selected rows... New

Catalog Item = Network Request

	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	visible/hide of ID field	(empty)		true	true	false	2026-10-02 02:37:44	100

1 to 1 of 1

Figure 5 - Catalog UI Policy 'visible/hide of ID field' linked to the item.

servicenow All Favorites History : Variable Set - Requester Information

Update Delete

* Title **Requester Information** Application Global

* Internal name requester_information Display title ☒

Order 100 Layout 2 Columns Wide, one side, then th

Type Single Row

Description

Update Delete

Variables (4) **Catalog UI Policies** Catalog Client Scripts Included In (1) Catalog Data Lookup Definitions

Order Search Actions on selected rows... New

Variable set = Requester Information

	Name	Type	Question	Order
☐	opened on behalf of	Reference	Opened on behalf of	

Figure 6 - Variable Set 'Requester Information' with four variables.

servicenow All Favorites History Admin : Catalog Client Scripts

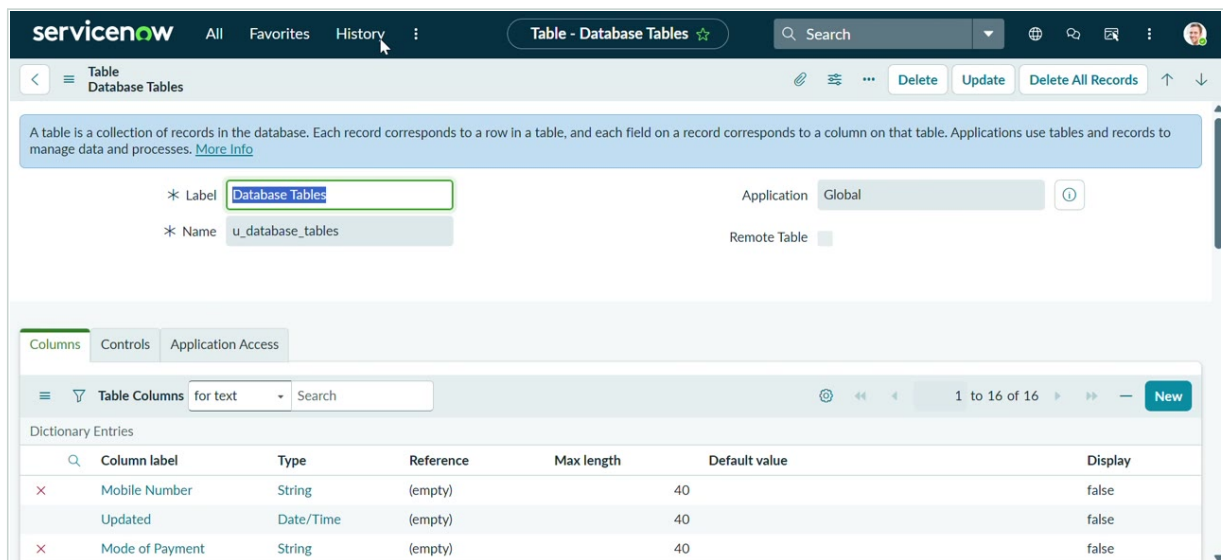
Name Search Actions on selected rows... New

All > Name >= network request

	Name	Active	Table	View	Type	Updated
☐	Search	Search	Search	Search	Search	Search
	onChange of Hide Mandatory	true			onChange	2022-11-20 06:07:10
	onChange of Mandatory Attachment	true			onChange	2022-11-20 06:07:21
	onChange Process engine	true			onChange	2022-11-20 06:04:05
	onLoad Process engine	true			onLoad	2022-11-28 01:09:52
	Populate applies to Item field	true			onLoad	2025-08-10 22:30:55
	Populate condition variable	true			onLoad	2021-08-03 01:20:41
	Populate Is Email & SMS active on load	true			onLoad	2023-03-22 12:24:15
	Populate Name with field	true			onChange	2025-04-13 03:18:08
	Populate Name with question text	true			onChange	2022-11-01 11:21:33
	Populate Prediction fields on load	true			onLoad	2022-06-03 08:08:09
	Populate Roles and Hide Variables	true			onLoad	2010-03-16 08:05:33

1 to 20 of 81

Figure 7 - Catalog Client Scripts filtered for the Network Request item.



A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label: Database Tables
 * Name: u_database_tables
 Application: Global
 Remote Table: ☐

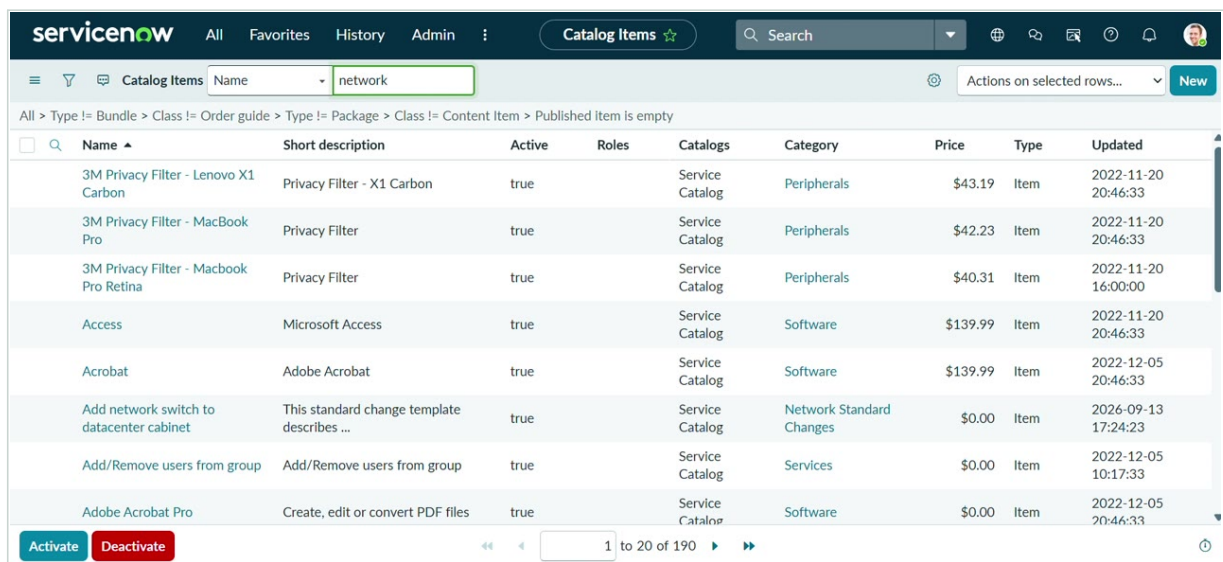
Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
×	Mobile Number	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
×	Mode of Payment	String	(empty)	40		false

Figure 8 - Custom table 'Database Tables' (u_database_tables) and its columns.



All > Type != Bundle > Class != Order guide > Type != Package > Class != Content Item > Published Item Is empty

	Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
	3M Privacy Filter - Lenovo X1 Carbon	Privacy Filter - X1 Carbon	true		Service Catalog	Peripherals	\$43.19	Item	2022-11-20 20:46:33
	3M Privacy Filter - MacBook Pro	Privacy Filter	true		Service Catalog	Peripherals	\$42.23	Item	2022-11-20 20:46:33
	3M Privacy Filter - Macbook Pro Retina	Privacy Filter	true		Service Catalog	Peripherals	\$40.31	Item	2022-11-20 16:00:00
	Access	Microsoft Access	true		Service Catalog	Software	\$139.99	Item	2022-11-20 20:46:33
	Acrobat	Adobe Acrobat	true		Service Catalog	Software	\$139.99	Item	2022-12-05 20:46:33
	Add network switch to datacenter cabinet	This standard change template describes ...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-09-13 17:24:23
	Add/Remove users from group	Add/Remove users from group	true		Service Catalog	Services	\$0.00	Item	2022-12-05 10:17:33
	Adobe Acrobat Pro	Create, edit or convert PDF files	true		Service Catalog	Software	\$0.00	Item	2022-12-05 20:46:33

Activate Deactivate 1 to 20 of 190

Figure 9 - Catalog Items list filtered by 'network'.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Single, standard intake for network requests	Fulfilment automation level is still 'Unspecified'
Fewer incomplete requests because of mandatory fields and validation	Requires a ServiceNow instance and admin skills to maintain
Reusable variable set speeds up building new items	Item description is empty, so form guidance is limited
Low-code configuration - easy to change	Older shared client scripts may conflict and need review
Trackable request numbers improve visibility	No approvals or notifications configured yet
Role-based access keeps the service secure	Depends on platform licensing and availability

9. CONCLUSION

The project delivers a structured Network Request item in the ServiceNow Service Catalog that replaces informal requests with a guided, validated and trackable process. By combining catalog variables, a reusable Requester Information variable set, a Catalog UI Policy and Catalog Client Scripts, the form collects complete information on the first attempt, and the process engine takes over once the request is submitted. The solution shows how low-code configuration on ServiceNow can reduce manual effort and delay in handling network standard changes, and it provides a base that can grow into full automation.

10. FUTURE SCOPE

- Add approval flows (for example, network manager approval) and automated fulfilment with Flow Designer.
- Integrate with network devices or the CMDB to validate and apply changes automatically.
- Send email / SMS notifications at each status change.
- Create dashboards and reports on request volume, turnaround time and SLA.
- Add a Service Portal widget and mobile-friendly experience, with an informative item description and search tags.
- Build Automated Test Framework (ATF) tests for regression testing.
- Extend the same pattern to other request types such as firewall, VPN and DNS.

11. APPENDIX

Source Code

ServiceNow configuration is mostly point-and-click. Below is an **illustrative** Catalog Client Script of the type used in the project (replace with your actual script from the Catalog Client Scripts list).

```
// Type: onChange Variable name: opened_on_behalf_of (illustrative example)
function onChange(control, oldValue, newValue, isLoading, isTemplate) {
    if (isLoading || newValue === '') {
        return;
    }
    // Show the ID field only when a requester is selected
    g_form.setDisplay('id_field', true);
    g_form.setMandatory('id_field', true);
}
```

Dataset Link

Not applicable - the project uses ServiceNow platform tables, with no external dataset.

GitHub & Project Demo Link

GitHub repository	[Paste your GitHub link here]
Project demo video	Automated Network Request Management (screen recording, about 3 min 27 s) - [Paste link here]
ServiceNow instance	[Instance URL, e.g. https://devXXXXXX.service-now.com]